

# S TAMANNA SHAHIN

Guwahati, India 781022 | +91 7002865039 | tamanna.shahin1993@gmail.com | LinkedIn: Tamanna Shahin

## Professional summary

Senior GIS Analyst with 7+ years of expertise in geospatial analysis, remote sensing, and spatial data management across land administration, infrastructure, forest resource conservation, and geological surveys. Currently serving in the Department of Land Records & Surveys, Government of Assam, integrating Python automation with GIS platforms to modernize cadastral workflows, build enterprise databases, and optimize survey operations. Proficient in Esri and open-source technologies for large-scale geospatial data processing, dispute resolution, and decision-support systems. Recognized for delivering statewide training, quality assurance frameworks, and cross-departmental collaboration that drive data accuracy and operational efficiency.

## Skills

|                                                                                                                                                                                                           |                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>GIS Platforms:</b> ArcGIS Pro, ArcGIS Desktop, ArcGIS Online, ArcGIS Enterprise, Survey123, QGIS                                                                                                       | <b>Remote Sensing:</b> ERDAS Imagine, ENVI, Google Earth Engine                             |
| <b>Programming &amp; Automation:</b> Python (ArcPy, GeoPandas, Rasterio, Shapely, PyProj, OpenCV), R (spatial packages), SQL                                                                              | <b>Web &amp; Mobile Mapping:</b> ArcGIS Field Maps, Collector, Survey123, Trimble Field App |
| <b>GPS &amp; Survey Equipment:</b> Trimble GNSS, RTK Rovers                                                                                                                                               | <b>Data &amp; Project Tools:</b> Metadata Management, Google Earth Pro, JIRA, Power BI      |
| <b>Domain Expertise:</b> Land Records & Cadastral Mapping, Forest Resource Assessment, Geological Surveys, Infrastructure Planning, Spatial Data QA/QC, Cartographic Visualization, Training & Mentorship |                                                                                             |

## Professional Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>GIS Lead Analyst</b><br><b>Directorate of Land Records &amp; Surveys</b> – Guwahati, Assam, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10/2023 to Current |
| <ul style="list-style-type: none"><li>Automated extraction, validation, and topology correction for 50,000+ land parcels, reducing manual processing time by 75%.</li><li>Directed Survey Settlement Operations across 2,500+ sq. km covering 400+ villages, updating ownership records using drones and ground validation.</li><li>Established and managed an enterprise geodatabase consolidating 5+ million records of village boundaries, classifications, and ownership layers.</li><li>Produced 20+ technical SOPs and weekly reports, standardizing state-level geospatial workflows.</li><li>Deployed 30+ RTK-enabled DGPS Rovers (DA2) for precision data capture across low-connectivity regions.</li><li>Collaborated with 100+ Revenue officials, 50+ village administrators, and multiple legal teams to resolve cadastral disputes using spatial models.</li><li>Delivered 10+ QA/QC training sessions and facilitated statewide cadastral workshops impacting 200+ technical staff.</li></ul> |                    |
| <b>GIS Analyst</b><br><b>SBL Knowledge Services Pvt. LTD</b> – Guwahati, Assam, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 09/2022 to 10/2023 |
| <b>Project 1: Assam Project on Forest and Biodiversity Conservation (APFBC)</b> <ul style="list-style-type: none"><li>Programmed Landsat/Sentinel classifiers in Python achieving 94% accuracy in detecting deforestation over 1,000+ sq. km forest areas.</li><li>Built FMIS dashboards integrating 50+ datasets, enabling decision-making for conservation planning and carbon stock assessments.</li><li>Conducted field surveys across 30+ sites, recording species mix, canopy cover, biomass, and carbon content.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |

**Project 2: Geological Survey of India – Shillong, Meghalaya, India**

- Digitized and georeferenced 5,000+ geological maps and datasets for subsurface modeling.
- Applied geomorphological and aeromagnetic spatial analysis across 2,200 sq. km terrain for mineral exploration.
- Executed QA/QC validation of datasets in line with national geoscience standards, ensuring compliance for 100+ deliverables.

**GIS Land Consultant**

07/2018 to 07/2022

**WSP** – Noida, Uttar Pradesh, India

- Managed parcel mapping and topology validation for high-profile projects including High Speed Railway (Phases 1 & 2) and Heathrow Airport, covering 1,000+ km of corridors.
- Created ArcPy automation tools that optimized routes and modeled travel times, reducing manual workload by 60% in the Arundel Bypass project.
- Produced DEMs and terrain overlays for 10+ engineering simulations, supporting acquisition and design planning.
- Integrated GPS/GNSS data from 20+ survey teams, ensuring accuracy in field-to-office workflows.
- Supervised survey coordination across 50+ field deployments, aligning data delivery between UK and India stakeholders.

**Education**

|                                                                                                      |         |
|------------------------------------------------------------------------------------------------------|---------|
| <b>Post Graduate Diploma:</b> Remote Sensing & GIS<br><b>Jamia Millia Islamia</b> - New Delhi, India | 06/2018 |
| <b>Master of Science:</b> Geography<br><b>Bangalore University</b> - Bengaluru, India                | 05/2017 |
| <b>Bachelor of Arts:</b> Geography<br><b>Miranda House (Delhi University)</b> - New Delhi, India     | 06/2014 |

**Certifications**

- AI Tools & ChatGPT – Be10x/IIT Kharagpur
- Power BI – Office Masters
- ENVI Analytics – Esri India
- Erosion Risk Assessment – Jamia Millia Islamia
- Remote Sensing & Traffic Analysis – UGIT Conference

**Languages**

|                                                                                                                           |                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| English                                                                                                                   | German                                                                                                                    |
| <div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div> | <div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div> |
| Bilingual or Proficient (C2)                                                                                              | Beginner (A1)                                                                                                             |